MAZDAK GHASEMI TOOTKABONI

GIS Systems Analyst & Developer | Field-to-Back-Office Esri Solutions | Process Automation

Open to relocation | 0402 347 929 | Mazdak.gh1995@gmail.com

LinkedIn | Portfolio | GitHub | **Full Australian work rights**

PROFESSIONAL SUMMARY

GIS Systems Analyst & Developer and Esri Enterprise specialist, currently building the **City of Tea Tree Gully's works management solution** on the ArcGIS platform. Hands-on experience designing, deploying and automating works management and field-data-collection solutions across Victorian and South Australian local government and state agencies — bridging field operations to back-office systems with **ArcGIS Enterprise, Experience Builder, ArcGIS Dashboards, Survey123 and Field Maps**, combined with strong process automation (FME, Python, SQL, PostGIS) and integration experience across enterprise asset management platforms (**Confirm, Assetic**). Master of Spatial Engineering from the University of Melbourne. Builds production geospatial solutions from requirements to deployment with one consistent approach: automate the deterministic parts of a workflow, make every decision traceable, and keep people in control of the judgement calls — delivering well-documented systems that field crews and back-office teams actually use.

SELECTED PROJECTS DEMONSTRATING SOLUTION-DEVELOPMENT CAPABILITY

Vicmap M1 Validation Pipeline — Council property-update automation, open source · github.com/Maz2580/vicmap-m1-ai-pipeline

Built at Greater Shepparton to automate the fortnightly M1 cycle end-to-end: inbox monitoring for the Vicmap delivery, FME reload workspaces, Pozi Connect M1 generation, then validation of every row against the rates system (SQL Server) and Vicmap layers in ArcSDE. Deterministic rules settle clear cases; a language model reviews only ambiguous rows with strict-schema output; each row gets KEEP/REJECT with a reason and confidence for officer sign-off. **Read-only database access by design · published security audit with critical findings fixed · CI on Windows & Linux · reusable by any Victorian council.**

UMAMI — University of Melbourne · Urban digital twin & GeoAI platform · geollm.idigitaltwin.org/UMAMI

Designed and built a digital twin with ten studios on a CesiumJS 3D globe: a multi-agent cadastre assistant over **4.29M Victorian parcels**, U-Net heat and flood surrogates of SOLWEIG and HEC-RAS served via ONNX, live hazard intelligence from 10+ feeds with a 17-tool assistant, AI change detection and planning-compliance auditing. FastAPI, React, PostGIS/pgvector, multi-provider LLMs with MCP, Prometheus/Grafana monitoring and GitLab CI/CD.

Terrascout — Geospatial data scoping platform · terrascout.app

Architected an end-to-end commercial platform that turns a plain-English project brief into ranked satellite, aerial and LiDAR datasets with feasibility checks: FastAPI back end, React/TypeScript front end, PostGIS + pgvector, REST API, a CLI published on PyPI and an MCP server for AI assistants. Rules and embeddings run first and the LLM is called only below a confidence threshold; every AI agent has a deterministic fallback. **24 providers / 32 sensors · 46-command CLI · 7-tool MCP server · hardened SQL sandbox.**

UrbanMind — Urban scenario planning platform for Australian practice · urbanmind.terrascout.app

Planning Check against Clause 55, 52.06 car parking and VicSmart routes, plus 25 analysis models — eleven rebuilt on **79 cited Victorian and Australian sources** — drawing on ~25 government data providers (ABS, Vicmap, PTV GTFS, EPA, Melbourne Water). IFC/DXF design import, SketchUp and Blender connectors, a 27-tool AI agent and MCP server; a PostGIS cache made map layers 75–430× faster. Deployed behind Cloudflare Tunnel with self-hosted CI.

GeoSpark — Open-source spatial AI library on PyPI · github.com/Maz2580/geospark

Designed a protocol and MCP tools that expose deterministic geodesic, topology, routing and terrain computation to any language model, with agents for multi-step analysis; every answer carries its source and licence, and runs can be replayed. **v0.9 on PyPI · 38 operations · hosted 18-tool MCP server in the MCP Registry · 1,000+ tests · Apache 2.0.**

PROFESSIONAL EXPERIENCE

Systems Analyst / Developer (GIS) | City of Tea Tree Gully

Sep 2026 – Present | Adelaide, SA | Remote

- Developing and delivering the Council's **works management solution** across the City Operations portfolio — geospatial digital works-solutions that lift process efficiency and effectiveness.

- Building applications across the works lifecycle: **proactive work assignment against assets, crew scheduling, field data capture, contractor work request dashboards** and interactive geospatial map views of all works.
- Gathering requirements with business stakeholders and translating them into technical solutions on the **ArcGIS/Esri platform**; designing applications for in-house and external users.
- Supporting integrations with external systems via **APIs, web services and middleware platforms**; leading design of dashboard reporting that gives stakeholders actionable, data-driven insight.
- Creating design specifications, technical diagrams and user guides; troubleshooting ArcGIS web applications; coordinating testing of new features before release and delivering user training.

Data Engineer | CSDILA, University of Melbourne

Dec 2024 – Present | Melbourne, VIC | Remote

- Design and build **UMAMI**, the University's urban digital twin — multi-agent cadastre assistant over 4.29M parcels, heat and flood surrogates, live hazard intelligence and AI change detection, from architecture to production.
- Architect and automate end-to-end geospatial data workflows using **n8n, Python, SQL and PostGIS**, reducing manual intervention and improving the reliability of spatial analytics.
- **Dockerise APIs, schedulers and UIs with Compose; implement versioned, repeatable releases via GitLab CI/CD** and container registry; deploy on Google Cloud Platform with secrets in masked CI variables and least-privilege service accounts.
- Design JSON/GeoJSON API contracts and authentication patterns; build Python back ends and React front ends serving spatial features and analytics to research and industry users.
- Develop and govern geospatial metadata, data catalogues and custom schemas for data quality, interoperability and reproducibility.
- Deliver geospatial inputs to projects including Arup, Venice Biennale: Song of the Cricket and UEDLAB — processing satellite and field data (Google Earth Engine, Landsat, Sentinel-2), applying supervised/unsupervised classification (Random Forest, K-Means), and producing change detection, land cover and habitat mapping outputs.
- Train and serve ML models (U-Net surrogates, fine-tuned building detectors) via ONNX; deploy IoT integration via **MQTT** with ventilation, CO_2 and privacy-preserving occupancy analysis.

GIS & Assets Officer | Greater Shepparton City Council

Jun 2025 – Sep 2026 | Shepparton, VIC | On-site

- Administered Council's enterprise GIS environment across **ArcGIS Pro, ArcGIS Enterprise, ArcGIS Server and Geocortex** — governing metadata standards, schema/version control practices, test plans and cataloguing for reliable, auditable services.
- **Automated the fortnightly M1 property-update cycle end-to-end** with FME and Python — rules-first validation against the rates system and Vicmap, language-model review of ambiguous rows only, read-only database access and officer sign-off; later open-sourced for other Victorian councils.
- Linked the **Content Manager** records system to GIS — extracting record numbers from capital works exports and matching documents to ArcSDE layers through FME and Python.
- Managed council infrastructure assets — road surfaces and base, footpaths, shared paths, drainage (pits and pipes) and kerbs — across the full lifecycle in **Confirm asset management software**.
- Cross-validated records between Confirm and the corporate GIS database, correcting geometries and resolving discrepancies to maintain a single source of truth.
- Built reproducible feature sets for asset failure, degradation and lifecycle modelling — combining condition proxies, age/usage attributes, environmental context and topology to support evidence-based prioritisation.
- Developed and maintained **web map applications** and analytical maps for capital works planning, work programs, stakeholder briefings and operational reporting.
- Processed property and subdivision changes via the Victorian Edit System (VES); partnered with planning, engineering, capital works and property teams on spatial analysis and troubleshooting; applied least-privilege access for publishing and data services.

GIS Officer | Mansfield Shire Council

Sep 2024 – Jun 2025 | Mansfield, VIC | Hybrid

Sole GIS professional — owned the full GIS function including vendor relationships, software procurement and subscription administration.

- Owned end-to-end GIS service delivery for planning, rates, property and asset teams as the sole spatial professional — managing day-to-day spatial support, vendor engagement and procurement decisions.
- **Designed and automated ETL pipelines using Python, SQL and PostGIS** to validate, transform and load multi-source spatial datasets into the central GIS database.
- Consolidated and reconciled fragmented spatial datasets — ensuring data integrity, attribution, geometry quality, metadata and QA compliance across council systems.

- Conducted quality assessment, metadata management and routine updates for new and existing property and asset data within the GIS system.
- Integrated GIS workflows with **POZI Connect, SynergySoft, MapInfo Pro, MBS Premium and Assetic** — improving consistency between spatial, financial and asset records.
- Engaged with planning, rates and asset management teams to interpret user requirements, **document procedures** and communicate data updates across the organisation.

Research Assistant | CSDILA, University of Melbourne

Dec 2023 – Nov 2024 | Melbourne, VIC | Remote

- Conducted research on AI-driven generative urban design — applying advanced spatial analysis, modelling and integration techniques.
- Advanced spatial ontologies and 3D cadastre processing — contributing to structured, accurate and reliable geospatial data management.
- Integrated geospatial datasets into **GeoServer** and Digital Twin visualisation environments — improving accessibility for research and industry stakeholders.
- Cleaned, corrected and harmonised diverse spatial datasets through geometry correction, coordinate system adjustments and consistent styling to support analytical outputs.

Research Assistant (GIS Analyst) | Curtin University

May 2024 – Sep 2024 | Perth, WA | Remote

- Delivered geospatial analytics, predictive modelling and scenario planning enhancing the Envision Tomorrow Australia (ET-Oz) sustainable planning model using **ArcGIS Pro and ArcGIS Online**.
- Customised spatial datasets for Australian urban contexts; harmonised diverse datasets to improve model representation and analytical reliability.
- Collaborated across research and planning contexts to ensure spatial data accuracy, consistency and actionable insights from complex datasets.

GIS Analyst | Parks Victoria

May 2022 – May 2024 | Melbourne, VIC | Hybrid

- **Designed and deployed integrated field-to-back-office workflows using ArcGIS Online, ArcGIS Enterprise, Field Maps and Survey123** — supporting damage assessments across Victorian flood- and fire-affected regions, with offline sync and real-time leadership dashboards.
- Built and governed a dedicated **GIS database for disaster recovery** — incorporating field updates, validation and integrity controls across affected regions.
- Generated **ArcGIS Dashboards** and analytical reports — improving stakeholder communication and operational coordination across recovery, planning and operations functions.
- Led geospatial delivery for emergency response and recovery programs — including spatial routing, impact mapping and resource allocation supporting multi-disciplinary recovery teams.
- Managed complex spatial data workflows using **FME, ArcGIS Pro and ArcGIS Enterprise** — maintaining accurate GIS databases critical for rapid emergency response and logistical planning.
- Managed user access, secure data sharing and field tool deployment — ensuring recovery teams had reliable access to the right mapping and analysis tools.
- **Trained recovery team members on Esri field collection tools (Field Maps, Survey123)** — building organisational capability in mobile GIS and standardised spatial data capture.

Spatial Data Analyst | HUMAYA Automation

Mar 2022 – May 2022 | Melbourne, VIC | Remote

- Designed spatial databases (Esri geodatabases, SQL Server) supporting infrastructure placement, risk assessment and operational decision-making.
- Conducted spatial analysis to identify strategic infrastructure locations and landslide risk zones — enabling proactive planning and mitigation.
- Integrated geospatial datasets with IoT solutions for smart metering, precision agriculture, and logistics monitoring, tracking and analysis.
- Maintained GIS, 3D and GPS datasets for corporate databases; liaised with staff to scope spatial data capture needs and recommend appropriate hardware and workflows.

GIS Analyst | Avale Enterprises

May 2021 – Feb 2022 | Melbourne, VIC | Remote

- Improved spatial dataset accuracy through routing optimisation, geometry correction and data validation using **ArcGIS Pro and MapInfo**.
- Produced high-quality cartographic outputs and spatial visualisations using optimised symbology and **Portal for ArcGIS** — improving user experience and decision-making.

- Restructured geospatial databases and **ArcGIS Online administrative workflows** — improving accessibility, compliance and operational consistency.
- Validated spatial data quality and improved visualisation standards to support clearer interpretation and decision-making.

Site Supervisor / Construction Site Manager | **SSS Steel Structures & Construction**

Oct 2016 – Apr 2021 | Guilan Province, Iran | On-site

- Held site supervision and construction site management responsibilities across civil and steel structure projects — coordinating personnel, subcontractors, materials, schedules, inspections and quality assurance processes.
- Worked as technical advisor across construction sites — supporting contract documentation, regulatory compliance, stakeholder coordination, resource planning and project delivery within scope and timeline.

EDUCATION

Master of Spatial Engineering | University of Melbourne | Feb 2022 – Dec 2024

Capstone: Digital Twin with IoT for real-time indoor and outdoor environmental monitoring (LoRaWAN, MQTT, four-layer architecture)

Master of Petroleum Engineering | Azad University, Central Tehran Branch | 2017 – 2020

Bachelor of Civil Engineering | Azad University, Bandar-e-Anzali Branch | 2013 – 2017

TECHNICAL SKILLS

Esri & ArcGIS Stack: ArcGIS Pro, ArcGIS Enterprise, ArcGIS Server, ArcGIS Online, Portal for ArcGIS, Experience Builder, ArcGIS Dashboards, Survey123, Field Maps, Workforce, Web AppBuilder, Geocortex/VertiGIS, Arcade, ArcGIS API for Python; sharing, roles and security

Asset & Works Management Integration: Confirm, Assetic, Content Manager, POZI Connect, SynergySoft, MapInfo Pro, MBS Premium, Victorian Edit System (VES)

Automation, ETL & Databases: FME Form Advanced, FME Flow, Python, SQL, PostgreSQL/PostGIS, SQL Server/ArcSDE, Oracle, n8n, GeoPandas, Shapely; schema governance, metadata management, QA/QC frameworks

Solution Development: JavaScript/TypeScript, HTML, CSS, React, FastAPI, Flask, REST/GeoJSON APIs, Docker/Compose, GitLab CI/CD, GitHub Actions, Linux, Nginx, automated testing, technical documentation

Cloud, AI & Integration: Google Cloud Platform, Cloudflare, MQTT/IoT dashboards, LLM integration and MCP tool servers

Spatial Data Science (supporting): Google Earth Engine, Sentinel-2/Landsat, supervised/unsupervised classification, change detection, remote sensing

CERTIFICATIONS

79+ verified credentials across Esri GIS, FME, cloud, AI/ML and project management. Highlights: **FME Form Advanced** (Safe Software) · **Esri Spatial Data Science** · **ArcGIS API for Python series** (Esri) · **Google Project Management Professional** · **ITIL Foundation** · **Generative AI Leader** (Google). Full list available via portfolio.